



Lecture 6: Continuous random variables

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1. Review
2. Continuous random variables
3. Cumulative distribution function
4. Normal and exponential RVs
5. Recitation
6. Summary



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- Total Expectation theorem

$$\mathbf{E}[X] = \mathbf{P}(A_1)\mathbf{E}[X | A_1] + \cdots + \mathbf{P}(A_n)\mathbf{E}[X | A_n]$$

- Multiple discrete RVs

$$p_{X,Y}(x, y) = \mathbf{P}(X = x \text{ and } Y = y)$$

- Properties of PMF:

$$- \sum_x \sum_y p_{X,Y}(x, y) = 1$$

$$- p_X(x) = \sum_y p_{X,Y}(x, y)$$

$$- p_{X|Y}(x | y) = \mathbf{P}(X = x | Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

$$- \sum_x p_{X|Y}(x | y) = 1$$

- Independence

$$p_{X,Y}(x, y) = p_X(x)p_Y(y)$$

- If X and Y are independent:

$$\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$$

$$\mathbf{E}[g(X)h(Y)] = \mathbf{E}[g(X)]\mathbf{E}[h(Y)]$$

$$\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$$

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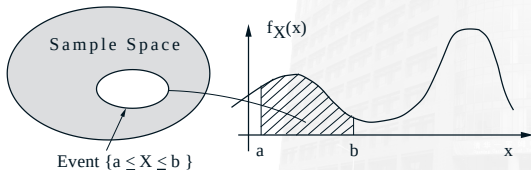
Definition of a continuous RV

A RV X is called **continuous** if there is a nonnegative function f_X , called the **probability density function of X** (PDF), such that

$$\mathbf{P}(X \in B) = \int_B f_X(x) dx$$

for every $B \subseteq \mathbb{R}$. In particular, the probability that X falls within an interval is

$$\mathbf{P}(a \leq X \leq b) = \int_a^b f_X(x) dx$$



Let X be a continuous RV with pdf f_X

- $f_X(x) \geq 0$ for all x (**nonnegativity**)
- $\int_{-\infty}^{\infty} f_X(x) dx = 1$ (**normalization**)
- $\mathbf{P}(x \leq X \leq x + \delta) = \int_x^{x+\delta} f_X(s) ds \approx f_X(x) \cdot \delta$ (**“mass per unit”**)

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Remark:

- For “nice” sets B ,

$$\mathbf{P}(X \in B) = \int_B f_X(x) dx$$

- Integrable: Riemann vs Lebesgue (Kolmogorov)

- Consider a RV X with PDF

$$f_X(x) = \begin{cases} \frac{1}{2\sqrt{x}}, & \text{if } 0 < x \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

- Even though $f_X(x)$ becomes infinitely large as x approaches zero, this is still a valid PDF, because

$$\int_{-\infty}^{\infty} f_X(x) dx = \int_0^1 \frac{1}{2\sqrt{x}} dx = \sqrt{x} \Big|_0^1 = 1$$

Definitions of mean and variance of a continuous RV

The **expected value** or **expectation** or **mean** of a continuous RV X is defined by

$$\mathbf{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

The **variance** of RV X is defined by

$$\text{var}(X) = \sigma_X^2 = \mathbf{E}[(X - \mathbf{E}[X])^2] = \int_{-\infty}^{\infty} (x - \mathbf{E}[X])^2 f_X(x) dx$$

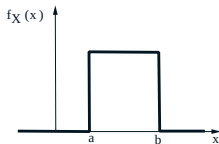
$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$$

- The mean of $g(X)$ satisfies the expected value rule

$$\mathbf{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

- Integration assumed to be “well-defined” (counterexample: Cauchy)

- **Uniform RV:** Consider a RV X that takes value in an interval $[a, b]$ with PDF



- Mean and variance:

$$\mathbf{E}[X] = \int_{-\infty}^{\infty} x f(x) dx = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{a+b}{2}$$

$$\mathbf{E}[X^2] = \int_a^b \frac{x^2}{b-a} dx = \frac{a^2 + ab + b^2}{3}$$

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = \frac{(b-a)^2}{12}$$

- **Exponential RV:** Consider an exponential RV X that has a PDF of the form

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

- Mean and variance

$$\mathbf{E}[X] = \int_0^{\infty} x \lambda e^{(-\lambda x)} dx = (-x e^{-\lambda x}) \Big|_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx = \frac{1}{\lambda}$$

$$\mathbf{E}[X^2] = \int_0^{\infty} x^2 \lambda e^{(-\lambda x)} dx = (-x^2 e^{-\lambda x}) \Big|_0^{\infty} + \int_0^{\infty} 2x e^{-\lambda x} dx = \frac{2}{\lambda^2}$$

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = \frac{1}{\lambda^2}$$

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- **Motivation:** to describe both **discrete** and **continuous** RVs with a single mathematical concept (PMF vs PDF)
- **Intuition:** Define the CDF to “accumulate” probability “up to” the value x .
- Consider event $\{X \leq x\}$

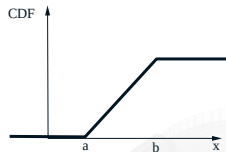
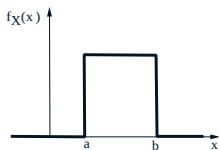
Definition of CDF

The CDF F_X of a RV X is defined by

$$F_X(x) = \mathbf{P}(X \leq x) = \begin{cases} \sum_{k \leq x} p_X(k), & \text{discrete} \\ \int_{-\infty}^x f_X(t) dt, & \text{continuous} \end{cases}$$

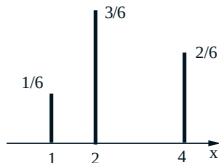
- Continuous RVs:

$$F_X(x) = \mathbf{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt$$

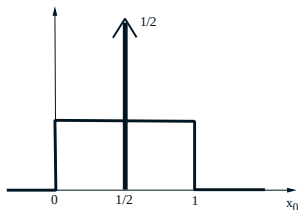


- Discrete RVs:

$$F_X(x) = \mathbf{P}(X \leq x) = \sum_{k \leq x} p_X(k)$$

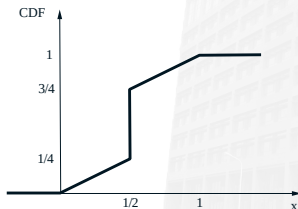


- Schematic drawing of a combination of a PDF and a PMF



- The corresponding CDF is a right continuous function

$$F_X(x) = \mathbf{P}(X \leq x)$$



Property of the CDF F_X :

- F_X is monotonically nondecreasing:

$$\text{if } x \leq y, \text{ then } F_X(x) \leq F_X(y)$$

- $F_X(x)$ tends to 0 as $x \rightarrow -\infty$ and to 1 as $x \rightarrow \infty$.
- If X is discrete, then $F_X(x)$ is a piecewise constant function of x
- If X is continuous, then $F_X(x)$ is a continuous function of x
- If X is continuous, then

$$F_X(x) = \int_{-\infty}^x f_X(t)dt, \quad f_X(x) = \frac{dF_X}{dx}(x)$$

- **Geometric RV:** Let X be a geometric RV with parameter p
- For $k = 1, 2, \dots$

$$\mathbf{P}(X = k) = p(1 - p)^{k-1}$$

- The CDF is given by

$$F_X(n) = \sum_{k=1}^n p(1 - p)^{k-1} = 1 - (1 - p)^n$$

- The PMF can be checked by

$$p_X(k) = F_X(k) - F_X(k - 1)$$

- Assume that each test takes one of the values from 1 to 10 with equal probability $1/10$ independently. What is the PMF of the maximum of three test scores?

$$X = \max\{X_1, X_2, X_3\}$$



- Assume that each test takes one of the values from 1 to 10 with equal probability $1/10$ independently. What is the PMF of the maximum of three test scores?

$$X = \max\{X_1, X_2, X_3\}$$

Solution:

- We have the CDF F_X

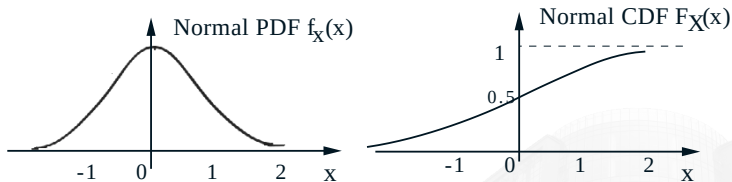
$$\begin{aligned} F_X(k) &= \mathbf{P}(X_1 \leq k, X_2 \leq k, X_3 \leq k) \\ &= \mathbf{P}(X_1 \leq k) \mathbf{P}(X_2 \leq k) \mathbf{P}(X_3 \leq k) \\ &= \left(\frac{k}{10}\right)^3 \end{aligned}$$

$$p_X(k) = F_X(k) - F_X(k-1) = \left(\frac{k}{10}\right)^3 - \left(\frac{k-1}{10}\right)^3, \quad k = 1, \dots, 10.$$

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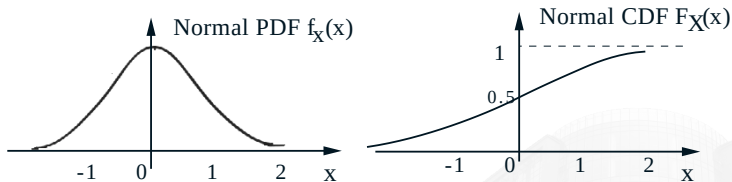


- Standard normal $N(0, 1)$: $f_X(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$



- $\mathbf{E}[X] =$
- $\text{var}(X) =$

- Standard normal $N(0, 1)$: $f_X(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$



- $\mathbf{E}[X] = 0$
- $\text{var}(X) = 1$

- General normal $N(\mu, \sigma^2)$: $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/2\sigma^2}$
- $\mathbf{E}[X] = \mu$ and $\text{var}(X) = \sigma^2$
- Let $Y = aX + b$
 - $\mathbf{E}[Y] =$
 - $\text{var}(Y) =$



- General normal $N(\mu, \sigma^2)$: $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/2\sigma^2}$
- $\mathbf{E}[X] = \mu$ and $\text{var}(X) = \sigma^2$
- Let $Y = aX + b$
 - $\mathbf{E}[Y] =$
 - $\text{var}(Y) =$
 - $Y \sim N(a\mu + b, a^2\sigma^2)$



Calculating normal probabilities

- No closed form available for CDF
- but there are tables (for standard normal)
- If $X \sim N(\mu, \sigma^2)$, then $\frac{X - \mu}{\sigma} \sim N(0, 1)$
- If $X \sim N(2, 16)$: $\mathbf{P}(X \leq 3) = \mathbf{P}\left(\frac{X-2}{4} \leq \frac{3-2}{4}\right) = \text{CDF}(0.25)$

	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767

- Revisit Exponential RV

- PDF: $f_X(x) = \lambda e^{-\lambda x}, x \geq 0$

- CDF: $\mathbf{P}(X \geq x) = \int_x^{\infty} \lambda e^{-\lambda x} dx = e^{-\lambda x}, x \geq 0$

- Revisit Exponential RV

- PDF: $f_X(x) = \lambda e^{-\lambda x}, x \geq 0$

- CDF: $\mathbf{P}(X \geq x) = \int_x^{\infty} \lambda e^{-\lambda x} dx = e^{-\lambda x}, x \geq 0$

- **Memoryless property:** for $x, c \geq 0$, we have

$$\mathbf{P}(X \geq c + x \mid X \geq c) = e^{-\lambda x} = \mathbf{P}(X \geq x)$$

- Probability of living another x time units is $e^{-\lambda x}$, no matter how long I have lived (“used as good as new”)

- Continuous-time analog of the geometric

- $\mathbf{P}(0 \leq X \leq \delta) \approx \lambda \delta$

- $\mathbf{P}(c \leq X \leq c + \delta \mid X \geq c) \approx \lambda \delta$

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1. Let Z be a continuous random variable with probability density function

$$f_Z(z) = \begin{cases} \gamma(1 + z^2), & \text{if } -2 < z < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) For what value of γ is this possible?
- (b) Find the cumulative distribution function of Z .

2. The taxi stand and the bus stop near Al's home are in the same location. Al goes there at a given time and if a taxi is waiting, (this happens with probability $2/3$) he boards it. Otherwise he waits for a taxi or a bus to come, whichever comes first. The next taxi will arrive in a time that is uniformly distributed between 0 and 10 minutes, while the next bus will arrive in exactly 5 minutes. Find the CDF and the expected value of Al's waiting time.

3. Let λ be a positive number. The continuous random variable X is called exponential with parameter λ when its probability density function is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{if } x \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find the cumulative distribution function (CDF) of X .
- (b) Find the mean of X .
- (c) Find the variance of X .
- (d) Suppose X_1 , X_2 , and X_3 are independent exponential random variables, each with parameter λ .
Find the PDF of $Z = \max\{X_1, X_2, X_3\}$.
- (e) Find the PDF of $W = \min\{X_1, X_2\}$.

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- Continuous random variables

$$\mathbf{P}(a \leq X \leq b) = \int_a^b f_X(x) dx$$

- Cumulative distribution functions

$$F_X(x) = \mathbf{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt$$

- Normal RV

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$$

- Exponential RV

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

- Readings: Section 3.1-3.3
- Assignment: Problem Set 6
- Next lecture: Multiple continuous random variables